

Name: Prashanth Raghavendra Rao

Task: K8s minikube

Date: 29/06/2025

Task Description:

Setup minikube at your local and explore creating namespaces (Go through official documentation).

```
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % brew install minikube

=> Downloading https://formulae.brew.sh/api/formula.json
=> Downloading https://formulae.brew.sh/api/cask.json
Warning: minikube 1.36.0 is already installed and up-to-date.
To reinstall 1.36.0, run:
  brew reinstall minikube
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % minikube start

🐳 minikube v1.36.0 on Darwin 15.5 (arm64)
🔧 Automatically selected the docker driver
👉 Using Docker Desktop driver with root privileges
🌟 Starting "minikube" primary control-plane node in "minikube" cluster
📡 Pulling base image v0.0.47 ...
📦 Downloading Kubernetes v1.33.1 preload ...
> preloaded-images-k8s-v18-v1...: 327.15 MiB / 327.15 MiB 100.00% 8.52 Mi
> gcr.io/k8s-minikube/kicbase...: 463.69 MiB / 463.69 MiB 100.00% 9.89 Mi
🔥 Creating docker container (CPUs=2, Memory=4000MB) ...
📦 Preparing Kubernetes v1.33.1 on Docker 28.1.1 ...
  ▪ Generating certificates and keys ...
  ▪ Booting up control plane ...
  ▪ Configuring RBAC rules ...
🔧 Configuring bridge CNI (Container Networking Interface) ...
📡 Verifying Kubernetes components...
  ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl get nodes

NAME        STATUS    ROLES    AGE   VERSION
minikube    Ready     control-plane   8s    v1.33.1
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl version --client

Client Version: v1.33.2
Kustomize Version: v5.6.0
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl get namespace

NAME                STATUS    AGE
default             Active   32s
kube-node-lease     Active   32s
kube-public         Active   32s
kube-system         Active   32s
prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl create namespace prashanth-namespace
namespace/prashanth-namespace created
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl get namespace

NAME                STATUS    AGE
default             Active   80s
kube-node-lease     Active   80s
kube-public         Active   80s
kube-system         Active   80s
prashanth-namespace Active   3s
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl run nginx --image=nginx --restart=Never -n prashanth-namespace
pod/nginx created
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl get pods -n prashanth-namespace

NAME    READY   STATUS    RESTARTS   AGE
nginx   0/1     ContainerCreating   0          10s
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ %
```

```
prashanthr — zsh — 195x64

Configuring bridge CNI (Container Networking Interface) ...
  Verifying Kubernetes components...
  Using image gcr.io/k8s-minikube/storage-provisioner:v5
  Enabled addons: storage-provisioner, default-storageclass
  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl get nodes

NAME        STATUS    ROLES    AGE   VERSION
minikube    Ready     control-plane  8s   v1.33.1
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl version --client

Client Version: v1.33.2
Kustomize Version: v5.6.0
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl get namespace
NAME                STATUS    AGE
default             Active   32s
kube-node-lease     Active   32s
kube-public         Active   32s
kube-system         Active   32s
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl create namespace prashanth-namespace
namespace/prashanth-namespace created
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl get namespace
NAME                STATUS    AGE
default             Active   89s
kube-node-lease     Active   89s
kube-public         Active   89s
kube-system         Active   89s
prashanth-namespace Active    3s
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl run nginx --image=nginx --restart=Never -n prashanth-namespace
pod/nginx created
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl get pods -n prashanth-namespace
NAME    READY   STATUS    RESTARTS   AGE
nginx   0/1     ContainerCreating   0          10s
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl delete namespace prashanth-namespace
namespace "prashanth-namespace" deleted
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl get namespace
NAME                STATUS    AGE
default             Active   4m28s
kube-node-lease     Active   4m28s
kube-public         Active   4m28s
kube-system         Active   4m28s
prashanth-namespace Active    3s
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % minikube stop

Stopping node "minikube" ...
Powering off "minikube" via SSH ...
1 node stopped.
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % minikube delete
Deleting "minikube" in docker ...
Deleting container "minikube" ...
Removing /Users/prashanthr/.minikube/machines/minikube ...
Removed all traces of the "minikube" cluster.
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ % kubectl get namespace
E0629 21:51:37.698152 7072 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get \"http://localhost:8080/api?timeout=32s\": dial tcp [::1]:8080: connection refused"
E0629 21:51:37.699840 7072 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get \"http://localhost:8080/api?timeout=32s\": dial tcp [::1]:8080: connection refused"
E0629 21:51:37.701315 7072 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get \"http://localhost:8080/api?timeout=32s\": dial tcp [::1]:8080: connection refused"
E0629 21:51:37.702736 7072 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get \"http://localhost:8080/api?timeout=32s\": dial tcp [::1]:8080: connection refused"
E0629 21:51:37.704156 7072 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get \"http://localhost:8080/api?timeout=32s\": dial tcp [::1]:8080: connection refused"
The connection to the server localhost:8080 was refused - did you specify the right host or port?
(base) prashanthr@Prashanth-MacBook-Pro-2 ~ %
```